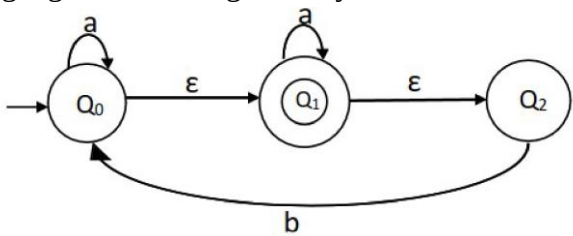
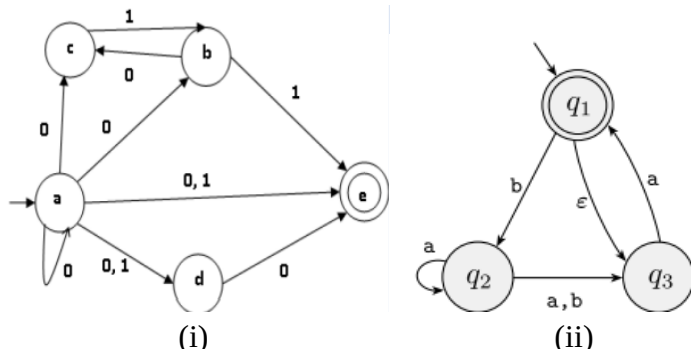


		ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		Assignment 1																																																			
Branch		CSE/CSIT		Programme		B.Tech																																																	
Course Name		Introduction to the Theory of Computation		Semester		5 th																																																	
Course Code		CSE3731		Academic Year		2025-26																																																	
ASSIGNMENT - 1																																																							
Submission due date:				08/11/2025																																																			
Learning Level (LL)		L1: Remembering		L3: Applying		L5: Evaluating																																																	
		L2: Understanding		L4: Analysing		L6: Creating																																																	
Q's	Questions					COs	LL																																																
1	a. Differentiate between complexity theory, computability theory, and automata theory in terms of their primary objectives, and explain how their frameworks collectively contribute to understanding the theoretical boundaries of computation. b. Define and distinguish the concepts of alphabet, string, and language within the context of automata theory, and illustrate each with suitable examples.					CO1	L1, L2																																																
2	a. Differentiate between the Kleene star closure (Σ^*) and Positive closure (Σ^+) with suitable examples. b. Let X be the set {1, 2, 3, 4, 5} and Y be the set {6, 7, 8, 9, 10}. The unary function $f : X \rightarrow Y$ and the binary function $g : X \times Y \rightarrow Y$ are described in the following tables. <table><tr><td>n</td><td>$f(n)$</td></tr><tr><td>1</td><td>6</td></tr><tr><td>2</td><td>7</td></tr><tr><td>3</td><td>6</td></tr><tr><td>4</td><td>7</td></tr><tr><td>5</td><td>6</td></tr></table><table><tr><td>g</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>1</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td></tr><tr><td>2</td><td>7</td><td>8</td><td>9</td><td>10</td><td>6</td></tr><tr><td>3</td><td>7</td><td>7</td><td>8</td><td>8</td><td>9</td></tr><tr><td>4</td><td>9</td><td>8</td><td>7</td><td>6</td><td>10</td></tr><tr><td>5</td><td>6</td><td>6</td><td>6</td><td>6</td><td>6</td></tr></table> i. What is the value of $f(2)$? ii. What are the range and domain of f ? iii. What is the value of $g(2, 10)$? iv. What are the range and domain of g ? v. What is the value of $g(4, f(4))$?					n	$f(n)$	1	6	2	7	3	6	4	7	5	6	g	6	7	8	9	10	1	10	10	10	10	10	2	7	8	9	10	6	3	7	7	8	8	9	4	9	8	7	6	10	5	6	6	6	6	6	CO1	L1, L2
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3	a. Using proof by construction, prove that for every finite undirected graph $G = (V, E)$, the sum of the degrees of all vertices in G is always an even number. b. Proof by contradiction that $\sqrt{2}$ is irrational. c. Let $C(n) = 1^3 + 2^3 + 3^3 + \dots + n^3$ be the sum of the first n cubes. Prove the following equalities by induction on n . $C(n) = \frac{1}{4}(n^4 + 2n^3 + n^2) = \frac{1}{4}n^2(n + 1)^2$					CO1	L1, L3																																																
4	Design the DFA's recognizing the following languages over the input alphabets $\Sigma = \{0, 1\}$. a) $\{w \mid w \text{ begins with a 1 and ends with a 0}\}$ b) $\{w \mid w \text{ contains the substring 0101 (i.e., } w = x0101y \text{ for some } x \text{ and } y)\}$ c) $\{w \mid w \text{ has length at least 3 and its third symbol is a 0}\}$ d) $\{w \mid w \text{ is any string except 11 and 111}\}$					CO2	L2, L3																																																

5	Let $D = \{w \mid w \text{ contains an even number of } a\text{'s and an odd number of } b\text{'s and does not contain the substring } ab\}$. a) Give a DFA with five states that recognizes D b) Define a regular expression that generates D.	CO2	L3, L4												
6	a. Design a NFA that accepts the language $L = \{w \mid w \text{ contains an even number of } 0\text{'s, or contains exactly two } 1\text{'s}\}$ over input alphabets $\Sigma = \{0, 1\}$ b. Convert the following ϵ-NFA to equivalent DFA and identify the language that is recognized by the DFA. 	CO2	L2, L3												
7	a. Design a DFA for the transition table as given below and determine the language that is recognized by the DFA. <table border="1" data-bbox="442 837 959 1061"><thead><tr><th>Present State</th><th>0</th><th>1</th></tr></thead><tbody><tr><td>$\rightarrow q_0$ (initial)</td><td>$\{q_0\}$</td><td>$\{q_1\}$</td></tr><tr><td>q_1</td><td>$\{q_1, q_2\}$</td><td>$\{q_2\}$</td></tr><tr><td>q_2</td><td>$\{q_1\}$</td><td>$\varnothing$</td></tr></tbody></table> b. Use the state construction (subset construction) method to convert the following two nondeterministic finite automata to equivalent deterministic finite automata. 	Present State	0	1	$\rightarrow q_0$ (initial)	$\{q_0\}$	$\{q_1\}$	q_1	$\{q_1, q_2\}$	$\{q_2\}$	q_2	$\{q_1\}$	$\varnothing$	CO2	L1, L3
Present State	0	1													
$\rightarrow q_0$ (initial)	$\{q_0\}$	$\{q_1\}$													
q_1	$\{q_1, q_2\}$	$\{q_2\}$													
q_2	$\{q_1\}$	$\varnothing$													
8	Let $\Sigma = \{a, b\}$. i) Write a regular expression to define the language consisting of all strings where the number of a's is even. ii) Write a regular expression to define language consisting of strings w such that, w contains only a's or only b's of length zero or more. iii) Write a regular expression to define the language consisting of all strings that end with either aa or bb. iv) Write a regular expression to define the language consisting of all strings that begin and end with the same symbol. v) Write a regular expression to define the language consisting of all strings that do not contain the substring ba. vi) Write a regular expression to define language consisting of strings w such that, w contains zero or more a's followed by zero or more b's.	CO3	L2, L3												

	vii) Write a regular expression to define the language consisting of all strings that contain at least one occurrence of the substring ab.		
9	a. Convert the following regular expressions to nondeterministic finite automata. i) $a(abb)^* \cup b$ ii) $((00)^*(11) \cup 01)^*$ iii) $(0^*1^*)^*000(0+1)^*$ b. Convert the following finite automata to regular expressions. (i) (ii)	CO2, CO3	L3
10	Use the pumping lemma to show that the following languages are not regular. a) $L1 = \{0^n 1^m 2^n \mid n, m \geq 0, n \text{ and } m \text{ are arbitrary integers}\}$ b) $L2 = \{w \mid w \text{ has an equal number of 0s and 1s}\}$	CO3	L2, L3

Course Outcomes	By the end of the course, through lectures, readings, home works, assignments, and exams, students will be able to:	
	CO1	Acquire knowledge of automata theory and develop proficiency in using mathematical terminology and proof strategies relevant to computational theory.
	CO2	Design deterministic and non-deterministic finite automata for regular languages.
	CO3	Develop and analyze regular expressions for regular languages, and establish their equivalence with finite automata through bidirectional conversion.
	CO4	Design context-free grammars and corresponding pushdown automata for the recognition of context-free languages.
	CO5	Design and analyze Turing machines for solving computational problems.
	CO6	Enhance proficiency in analyzing computational problems concerning decidability, undecidability, complexity classifications, and reducibility principles.

- ✓ Assignment scores/markings depend on neatness and clarity.
- ✓ Plagiarized assignments will be given a zero mark.
- ✓ Submit the hard copy of your assignment by the due date, i.e. **08.11.2025**
- ✓ Submit the assignment handwritten on A4 size papers and spirally bound to your ITC class teacher.
A front page must be present containing the details of the subject, the assignment and the student.